THE DIAGONAL OF A LOCALLY FINITE TYPE MORPHISM IS LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof treats both sentences in one environment; the second is derived from the first together with the definitions and the separatedness of the diagonal.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Stability of local finite typeness under base change. Lemma lemma-base-change-finite-type, used at proof lines 5858-5859. Makes X x_Y X of finite type over X.
- Permanence of local finite presentation over a locally finite type intermediate space. Lemma lemma-finite-presentation-permanence (5826-5833), used at proof lines 5860-5861. Gives local finite presentation of the diagonal.
- The diagonal of a morphism of algebraic spaces is separated. Lemma lemma-properties-diagonal (381-394), used at proof lines 5862-5864. Upgrades the local statement to the global one under quasi-separatedness.
- Quasi-separatedness of f as quasi-compactness of the diagonal. used at proof lines 5850-5852. Supplies the compactness half of the second statement.

Source reference. Original lemma 03XH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5148–5148. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05WT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7597–7597. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4570–4570. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S with diagonal $\Delta : X \to X \times_Y X$. If f is locally of finite type then Δ is locally of finite presentation. If f is quasi-separated and locally of finite type, then Δ is of finite presentation.*

Proof. Note that Δ is a morphism over X (via the second projection $X \times_Y X \to X$). Assume f is locally of finite type. Note that X is of finite presentation over X and $X \times_Y X$ is of finite type over X (by Lemma 03XH). Thus the first statement holds by Lemma 05WT. The second statement follows from the first, the definitions, and the fact that a diagonal morphism is separated (Lemma 03HK). □